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Roll No. _____

BS/Sem V/268

B. Sc. (Hons) Semester V Examination
2 0 1 5 – 1 6

PHYSICS

Paper No. BPT-505 : Electromagnetic Theory

Time : 3 hours

Full Marks : 70

(Write your Roll No. at the top immediately on the receipt of this question paper)

The figures in the right-hand margin indicate marks

Answer all questions

Mathematical symbols used in the text carry their usual meanings

1. Answer any four questions :

4×5

- (a) Define the curl of a vector field and explain the physical meaning associated with it.
- (b) Is $\vec{\nabla} \cdot (\vec{\nabla} \times \vec{v}) = 0$? How can one prove this using the theorems of vector calculus?
- (c) Show that the electric field of an infinite plane which carries a uniform surface charge is independent of radial distance.
- (d) What is Poynting vector? What role does it play in momentum conservation in electrodynamics?
- (e) Which TE modes will propagate in a hollow rectangular waveguide with dimensions $2.28 \text{ cm} \times 1.01 \text{ cm}$ if the driving frequency is $1.70 \times 10^{10} \text{ Hz}$?
- (f) What is skin effect? Calculate the skin depth of an EM wave at 100 MHz frequency in sea water, given that $\sigma = 5.0 \text{ mho/m}$, $\varepsilon = 6 \times 10^{-10} \text{ C}^2/\text{N} \cdot \text{m}^2$ and $\mu = \mu_0 = 4\pi \times 10^{-7} \text{ N/A}^2$.

— (2) —

- 2. (a) What are bound charges? Derive an expression for bound charge densities in terms of polarization vector. **6**
- (b) Write the Maxwell's equations in electro-statics in differential form and discuss the boundary conditions for tangential component of electric field across a surface. **6**

— (3) —

- 3. (a) Consider EM wave propagation along z -direction in a coaxial waveguide. For TEM mode the electric field amplitude is given as,

$$\vec{E}_0 = E_0 \frac{\hat{\rho}}{\rho}$$

State and explain the Gauss's law in magneto-statics. How this law will be modified if magnetic monopoles are detected by experiments? **6**

- (a) Show that for a hollow cylindrical guide for TE mode of wave propagation, the magnetic field component along the direction of propagation (say x) satisfies the following equation:

$$\left[\frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} + \frac{\omega^2}{c^2} - k^2 \right] B_x = 0$$

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- (b) Solve the above equation to find an expression for the cut-off frequency for the waveguide.

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ExternalPortals&AcademicResources:

- **ExaminationPortal:** <https://bhu-exam.vercel.app/>
- **StudentResourceCenter:** <https://quashan-me.vercel.app/>
- **AcademicPortfolio:** <https://me-aryan.vercel.app>